Fiscal Year:	FY 2010	**Task Last Updated:**	FY 11/18/2009
PI Name:	Volz, Martin Ph.D.		
Project Title:	Reduction of Defects in Germanium-Silicon (RDGS)		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	35812	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-02
Start Date:	03/06/2008	**End Date:**	03/05/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Szofran, Frank	**Contact Phone:**	256-544-7777
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
	NOTE: Original Principal Investigator was Dr. Frank Szofran, Feb 1998-Jan 2004; project continues with Dr. Martin Volz. It is well established that crystals grown without contact with a container have far superior quality to otherwise similar crystals grown in direct contact with a container. In addition to float-zone processing, detached-Bridgman growth is often cited as a promising tool to improve crystal quality, without the limitations of float zoning. In detached growth, most of the melt has contact with the crucible wall, but the crystal is separated from the wall by a small gap, typically 10-100 microns. A liquid meniscus extends between the top edge of the growing crystal and the crucible wall. Detached growth has been found to occur quite often during microgravity experiments and considerable improvements of crystal quality have been reported for those cases. However, no thorough understanding of the process or quantitative assessment of the quality improvements existed prior to the investigation. This project will determine the means to reproducibly grow Ge-Si alloys in the detached mode.		

Task Description:	Ge-Si alloys are of technological interest because with lower defect density materials it would be possible to make thermal neutron and x-ray optics, faster devices than with silicon, higher efficiency solar cells, and higher thermopower devices. Although these alloys are model materials in the sense that these results should be applicable to many other materials, the major technological significance is that the same production equipment that is currently used for silicon device manufacture can also be used for germanium-silicon devices. This means that using germanium-silicon instead of some other material to achieve improved device performance will save an immense amount of money in development and purchase of production equipment. Several tasks have been completed or are planned as part of the investigation: * measure the relevant material parameters such as contact angle, growth angle, surface tension, and wetting behavior of the GeSi-melt on potential crucible materials; * determine the mechanism of detached growth including the role of convection; * quantitatively determine the differences in defects and impurities among normal Bridgman, detached Bridgman, and float zone (FZ) growth; * investigate the influence of defined azimuthal or meridional flow due to rotating magnetic fields on the characteristics of detached growth; * control time-dependent Marangoni convection in the case of FZ-growth by the use of a rotating magnetic field to examine the influence on the curvature of the solid-liquid interface and the heat and mass transport; and * grow high quality GeSi single crystals with Si concentrations up to 10 at% and diameters up to 20 mm. Determine the necessary conditions for reproducibly achieving detachment of Ge-Si alloys (up to 10% Si) in the Bridgman growth method, and quantitatively compare the defect structures and impurity levels of detached Bridgman, normal Bridgman, and floating-zone grown materials.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this research will increase our understanding of the mechanisms which lead to crystal detachment and enhance our ability to control detachment both in microgravity and on Earth. Control of detachment will enable the growth of relatively large diameter crystals with orders of magnitude fewer defects than can be achieved with standard growth techniques. The improvement in material quality may lead to better performance for a variety of electronic devices.
Task Progress:	The thermal field of the Bridgman furnace used to grow detached germanium at MSFC was studied. The detailed heat transfer computations were performed using the CrysMAS code. This numerical study was done in collaboration with C. Stelian (West University of Timisoura, Romania) and J. Derby (University of Minnesota). The study examined the influence of thermal phenomena on crystal reattachment during detached Bridgman growth. Several results were drawn from this study. Consistent with the experimental results, it was found that the large wetting angle of germanium on boron nitride surfaces was an important factor to promote conditions for stable, detached growth. The computational results also indicated that the prevention of crystal reattachment to the crucible wall required the avoidance of any undercooling of the melt meniscus during the growth run. This could be accomplished by repositioning the ampoule with respect to the furnace zones. Specifically, it was predicted that by initiating growth much higher into the hot zone, favorable thermal conditions for continued detachment could be obtained, even for systems that did not exhibit large wetting angles. Essentially, crystal detachment could be maintained during growth as long as the solidification isotherm could be kept below the meniscus position. Control of the solid-liquid interface shape is important, both for bulk crystal growth in general and in detached growth. A typical Bridgman crystal growth system has a hot zone, with a temperature above the solidification point, a gradient zone, and a cold zone, with a temperature below the solidification point. We conducted numerical calculations to assess the effect of additional localized radial heating applied near the solid-liquid interface on the interface shape during Bridgman growth. System parameters examined include the ampoule, melt and crystal thermal conductivities, the magnitude and width of localized heating, and the latent heat of crystallization. Concave interface shapes, typical of semiconductor systems, could be flattened or made convex with localized heating. Although localized heating caused shallower thermal gradients ahead of the interface, the magnitude of the localized heating required for convexity was less than that which resulted in a thermal inversion ahead of the interface. A convex interface shape was most readily achieved with ampoules of lower thermal conductivity. Increasing melt convection tended to flatten the interface, but the amount of radial heating required to achieve a convex interface was essentially independent of the convection intensity. A review of some of the project results on detached Bridgman growth was highlighted in a feature article in the April, 2009 issue of the MRS Bulletin, the periodical magazine of the Materials Research Society. Key theoretical and experimental results were described. Theoretical meniscus shapes were calculated as a function of the pressure differential across the meniscus and the contact angle the meniscus makes with the ampoule wall. Two cases were considered. The first is when the sum of the growth angle and contact angle are less than 180 degrees and the second is when the sum of those angles is greater than 180 degrees. In the first case, the gap width is small for large positive pressure differential (pressure below the meniscus larger than above the meniscus), increases as the pressure differential decreases, and then decreases again as the pressure differential becomes increasingly negative. In the second case, a meniscus can only exist for negative pressure differentials and the gap width increases as the pressure differential approaches zero. In either case, detached growth can only occur when the meniscus is dynamically stable and when the gap width is larger than the size of the roughness of the ampoule. Experimental results included sessile drop measurements of wetting angles of germanium on different substrates. The highest wetting angles for germanium were measured at 172 degrees on a pyrolitic boron nitride substrate. During growth, there are several methods to generate and control a pressure differential across the meniscus. One of those methods involves changing the axial temperature gradient in a multizone furnace after melting but before growth. After the starting material is molten, the volume around and below the seed area is sealed off from the gas volume above the

	melt. The pressure above and below the melt can then be controlled by adjusting the furnace zone temperatures in these areas. By this method it was possible to grow completely detached germanium in boron nitride ampoules. Finally, it was noted that the etch pit density in detached grown germanium is 100 times less than the etch pit density of attached grown germanium.
Bibliography Type:	Description: (Last Updated: 07/09/2021)
Abstracts for Journals and Proceedings	Stelian C, Volz MP, Derby JJ. "Numerical optimization of the thermal field in Bridgman detached growth." Presented at the 17th American Conference on Crystal Growth and Epitaxy, Lake Geneva, WI, August 9-14, 2009. 17th American Conference on Crystal Growth and Epitaxy 2009. Abstract Book. , Aug-2009
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